

Econometrics Assignment 1

Subject code: DA242

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M.Sc. Big Data Analytics, CS Department
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1 Question 1

In each of the following data set:

- (a) Find the correlation coefficient between x and y and comment on it, and
- (b) Plot the scatter diagram and comment on it

1.1 Data set 1

y_1 :	0.8	1.2	1.5	1.8	2.2	2.3	2.6	3.1
x_1 :	1.7	2.7	3.6	4.6	5.7	6.7	8.1	12.0

Table 1: Expenditure on a commodity and income of 8 persons

Here, y_1 is per capita expenditure on a commodity and x_1 is per capita income.

Solution.

- (a) Given total number of points $n = 8$.

$$\bar{x}_1 = \frac{\sum x_{1,i}}{n} = \frac{1.7 + 2.7 + 3.6 + 4.6 + 5.7 + 6.7 + 8.1 + 12}{8} = \frac{45.1}{8} = 5.6375$$

$$\bar{y}_1 = \frac{\sum y_{1,i}}{n} = \frac{0.8 + 1.2 + 1.5 + 1.8 + 2.2 + 2.3 + 2.6 + 3.1}{8} = \frac{15.5}{8} = 1.9375$$

We know that the formula of covariance i.e.,

$$Cov(x_1, y_1) = \frac{\sum (x_{1,i} - \bar{x}_1)(y_{1,i} - \bar{y}_1)}{n}$$

and sample standard deviations of x_1, y_1 are

$$s_{x_1} = \sqrt{\frac{\sum (x_{1,i} - \bar{x}_1)^2}{n-1}}, \quad s_{y_1} = \sqrt{\frac{\sum (y_{1,i} - \bar{y}_1)^2}{n-1}}$$

respectively.

So, let us calculate each of them:

$$\begin{aligned} \sum (x_{1,i} - \bar{x}_1)^2 &= (1.7 - 5.6375)^2 + (2.7 - 5.6375)^2 + (3.6 - 5.6375)^2 \\ &\quad + (4.6 - 5.6375)^2 + (5.7 - 5.6375)^2 + (6.7 - 5.6375)^2 \\ &\quad + (8.1 - 5.6375)^2 + (12 - 5.6375)^2 \\ &= 77.03875 \end{aligned} \quad (1)$$

$$\begin{aligned} \sum (y_{1,i} - \bar{y}_1)^2 &= (0.8 - 1.9375)^2 + (1.2 - 1.9375)^2 + (1.5 - 1.9375)^2 \\ &\quad + (1.8 - 1.9375)^2 + (2.2 - 1.9375)^2 + (2.3 - 1.9375)^2 \\ &\quad + (2.6 - 1.9375)^2 + (3.1 - 1.9375)^2 \\ &= 4.03875 \end{aligned} \quad (2)$$

$$\begin{aligned} \sum (x_{1,i} - \bar{x}_1)(y_{1,i} - \bar{y}_1) &= (1.7 - 5.6375)(0.8 - 1.9375) + (2.7 - 5.6375)(1.2 - 1.9375) \\ &\quad + (3.6 - 5.6375)(1.5 - 1.9375) + (4.6 - 5.6375)(1.8 - 1.9375) \\ &\quad + (5.7 - 5.6375)(2.2 - 1.9375) + (6.7 - 5.6375)(2.3 - 1.9375) \\ &\quad + (8.1 - 5.6375)(2.6 - 1.9375) + (12.0 - 5.6375)(3.1 - 1.9375) \\ &= 17.10 \end{aligned} \quad (3)$$

Hence,

$$Cov(x_1, y_1) = \frac{\sum (x_{1,i} - \bar{x}_1)(y_{1,i} - \bar{y}_1)}{n-1} = \frac{17.10}{7} = 2.44$$

We also know that Pearson's correlation coefficient r is defined by

$$r = \frac{\sum (x_{1,i} - \bar{x}_1)(y_{1,i} - \bar{y}_1)}{\sqrt{\sum (x_{1,i} - \bar{x}_1)^2} \sqrt{\sum (y_{1,i} - \bar{y}_1)^2}}$$

As per the calculations made,

$$r = \frac{17.10}{\sqrt{77.04} \sqrt{4.04}} = 0.9699$$

$$\implies r^2 \approx 0.94$$

Hence, $r \approx 0.970$ is a very strong positive linear correlation between per-capita income and per-capita expenditure. About 94% of the variance in y_1 (expenditure) is linearly associated with x_1 (income) under this simple linear model.

- (b) **Comments:** The following scatter diagram shows a **strong positive linear relationship** between per capita income and per capita expenditure.

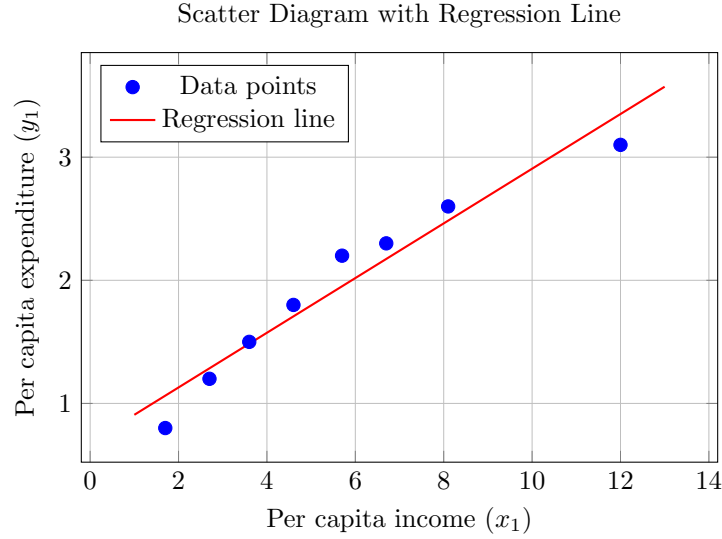


Figure 1: Scatter diagram of per capita income vs. expenditure with regression line

1.2 Data Set 2

y_2	70	65	90	95	110	115	120	140	155	150
x_2	80	100	120	140	160	180	200	220	240	260

Table 2: y_2 and x_2 values of 10 units in the sample

Solution.

- (a) Given total number of points $n = 10$.

$$\bar{x}_2 = \frac{\sum x_{2,i}}{n} = \frac{80 + 100 + 120 + 140 + 160 + 180 + 200 + 220 + 240 + 260}{10} = \frac{1700}{10} = 170$$

$$\bar{y}_2 = \frac{\sum y_{2,i}}{n} = \frac{70 + 65 + 90 + 95 + 110 + 115 + 120 + 140 + 155 + 150}{10} = \frac{1110}{10} = 111$$

We know that the formula of covariance i.e.,

$$\text{Cov}(x_2, y_2) = \frac{\sum (x_{2,i} - \bar{x}_2)(y_{2,i} - \bar{y}_2)}{n}$$

and sample standard deviations of x_2, y_2 are

$$s_{x_2} = \sqrt{\frac{\sum (x_{2,i} - \bar{x}_2)^2}{n-1}}, \quad s_{y_2} = \sqrt{\frac{\sum (y_{2,i} - \bar{y}_2)^2}{n-1}}$$

respectively.

So, let us calculate each of them:

$$\begin{aligned} \sum (x_{2,i} - \bar{x}_2)^2 &= (80 - 170)^2 + (100 - 170)^2 + (120 - 170)^2 + (140 - 170)^2 \\ &\quad + (160 - 170)^2 + (180 - 170)^2 + (200 - 170)^2 + (220 - 170)^2 \\ &\quad + (240 - 170)^2 + (260 - 170)^2 \\ &= 33000 \end{aligned} \tag{4}$$

$$\begin{aligned} \sum (y_{2,i} - \bar{y}_2)^2 &= (70 - 111)^2 + (65 - 111)^2 + (90 - 111)^2 + (95 - 111)^2 \\ &\quad + (110 - 111)^2 + (115 - 111)^2 + (120 - 111)^2 + (140 - 111)^2 \\ &\quad + (155 - 111)^2 + (150 - 111)^2 \\ &= 8890 \end{aligned} \tag{5}$$

$$\begin{aligned} \sum (x_{2,i} - \bar{x}_2)(y_{2,i} - \bar{y}_2) &= (80 - 170)(70 - 111) + (100 - 170)(65 - 111) \\ &\quad + (120 - 170)(90 - 111) + (140 - 170)(95 - 111) \\ &\quad + (160 - 170)(110 - 111) + (180 - 170)(115 - 111) \\ &\quad + (200 - 170)(120 - 111) + (220 - 170)(140 - 111) \\ &\quad + (240 - 170)(155 - 111) + (260 - 170)(150 - 111) \\ &= 16800 \end{aligned} \tag{6}$$

Hence, using the sample (n-1) denominator for covariance we get

$$\text{Cov}(x_2, y_2) = \frac{\sum (x_{2,i} - \bar{x}_2)(y_{2,i} - \bar{y}_2)}{n-1} = \frac{16800}{9} = 1866.66$$

We also know that Pearson's correlation coefficient r is defined by

$$r = \frac{\sum (x_{2,i} - \bar{x}_2)(y_{2,i} - \bar{y}_2)}{\sqrt{\sum (x_{2,i} - \bar{x}_2)^2} \sqrt{\sum (y_{2,i} - \bar{y}_2)^2}}$$

Substituting the computed sums,

$$r = \frac{16800}{\sqrt{33000} \sqrt{8890}} = \frac{16800}{\sqrt{33000 \cdot 8890}} \approx 0.9808473686$$

$$\Rightarrow r^2 \approx 0.96206$$

Hence, $r \approx 0.981$ indicates a **very strong positive linear correlation** between per-capita income (x) and per-capita expenditure (y_2). About 96.21% of the variance in y is linearly associated with x_2 under this simple linear model.

The sample standard deviations (using $n - 1$) are

$$s_x = \sqrt{\frac{33000}{9}} \approx 60.5530, \quad s_y = \sqrt{\frac{8890}{9}} \approx 31.4289.$$

The least-squares regression line parameters (using the usual formulas)

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{16800}{33000} = 0.5090909090 \dots$$

$$b_0 = \bar{y} - b_1 \bar{x} = 111 - 0.5090909090 \cdot 170 \approx 24.45454545$$

Therefore the fitted line is

$$\hat{y} = 24.454545 + 0.5090909, x.$$

- (b) **Comments:** The scatter diagram (below) shows a **strong positive linear relationship** between per capita income and per capita expenditure. The data points lie very near the fitted regression line, consistent with the high value of $r \approx 0.981$.

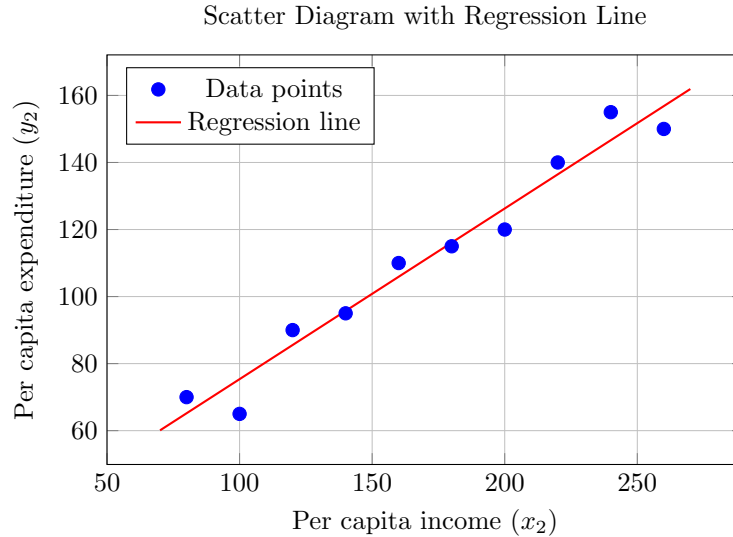


Figure 2: Scatter diagram of per capita income vs. expenditure with regression line

1.3 Data Set 3

y_3	256	414	634	778	1003	1839	2081	2423	2734	2914
x_3	1000	2000	3000	4000	5000	6000	7000	8000	9000	10000

Table 3: Weekly payments to the sales representatives (y_3) and sales of the sales representatives (x_3)

Here y_3 is the weekly payment to the sales representatives based on their sales (x_3).

Solution.

(a) Given total number of points $n = 10$.

$$\bar{x}_3 = \frac{\sum_{i=1}^{10} x_{3,i}}{n} = \frac{1000 + 2000 + 3000 + 4000 + 5000 + 6000 + 7000 + 8000 + 9000 + 10000}{10} = \frac{55000}{10} = 5500$$

$$\bar{y}_3 = \frac{\sum_{i=1}^{10} y_{3,i}}{n} = \frac{256 + 414 + 634 + 778 + 1003 + 1839 + 2081 + 2423 + 2734 + 2914}{10} = \frac{15076}{10} = 1507.6$$

We use the sample covariance and sample standard deviations:

$$Cov(x_3, y_3) = \frac{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)(y_{3,i} - \bar{y}_3)}{n - 1},$$

$$s_{x_3} = \sqrt{\frac{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)^2}{n - 1}}, \quad s_{y_3} = \sqrt{\frac{\sum_{i=1}^{10} (y_{3,i} - \bar{y}_3)^2}{n - 1}}.$$

Now compute the required sums (aggregated results):

$$\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)^2 = 82,500,000$$

$$\sum_{i=1}^{10} (y_{3,i} - \bar{y}_3)^2 = 9,071,166.4$$

$$\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)(y_{3,i} - \bar{y}_3) = 26,926,000$$

Hence the sample covariance is

$$Cov(x_3, y_3) = \frac{26,926,000}{9} = 2,991,777.77$$

Sample standard deviations:

$$s_{x_3} = \sqrt{\frac{8,25,00,000}{9}} \approx 3027.65 \quad , \quad s_{y_3} = \sqrt{\frac{90,71,166.4}{9}} \approx 1003.95$$

Pearson's correlation coefficient (sums form) is

$$r = \frac{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)(y_{3,i} - \bar{y}_3)}{\sqrt{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)^2} \sqrt{\sum_{i=1}^{10} (y_{3,i} - \bar{y}_3)^2}} = \frac{2,69,26,000}{\sqrt{8,25,00,000} \sqrt{90,71,166.4}} \approx 0.984$$

$$\implies r^2 \approx 0.968$$

Thus $r \approx 0.9843$ indicates a **very strong positive linear correlation** between the sales (x_3) and weekly payments (y_3). About 96.88% of the variation in y_3 is linearly associated with x_3 under this simple linear model.

The least-squares regression line (fitting y_3 on x_3) is obtained by

$$b_1 = \frac{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)(y_{3,i} - \bar{y}_3)}{\sum_{i=1}^{10} (x_{3,i} - \bar{x}_3)^2} = \frac{2,69,26,000}{8,25,00,000} = 0.32637576 \dots$$

$$b_0 = \bar{y}_3 - b_1 \bar{x}_3 = 1507.6 - 0.3263757576 \cdot 5500 \approx -287.4669$$

Therefore the fitted regression equation is (rounded)

$$\hat{y}_3 = -287.4667 + 0.326376 x_3.$$

- (b) **Comments:** The scatter diagram (below) shows a clear and **very strong positive linear relationship** between sales x_3 and weekly payments y_3 . The data points lie very close to the fitted regression line $\hat{y}_3 = -287.466667 + 0.326376 x_3$, which matches the high correlation $r \approx 0.9843$. The negative intercept is an extrapolation artifact outside the observed x_3 range and does not affect the slope's interpretation within the observed data.

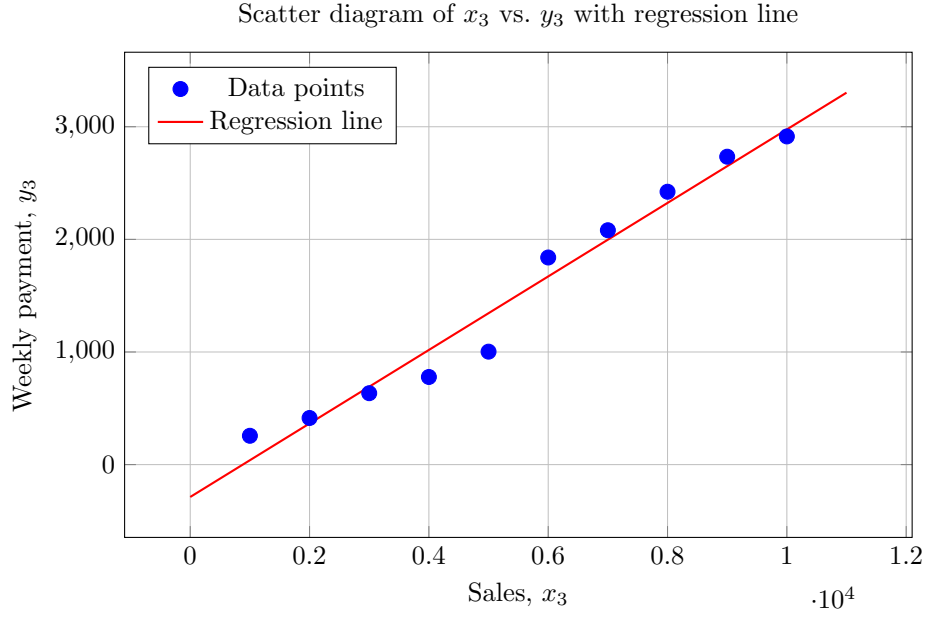


Figure 3: Scatter diagram of sales (x_3) vs. weekly payments (y_3) with fitted regression line

1.4 Data Set 4

Family No.	Income (x_4)	Consumption Expenditure (y_4)
1	80	70
2	98	76
3	105	91
4	115	100
5	125	105
6	135	113
7	145	122
8	155	120
9	165	146
10	175	135
11	185	147

Table 4: Weekly consumption expenditure and income of 11 families

Solution.

The correlation coefficient between x_4 and y_4 is given by:

$$r = \frac{\sum_{i=1}^n (x_{4i} - \bar{x}_4)(y_{4i} - \bar{y}_4)}{\sqrt{\sum_{i=1}^n (x_{4i} - \bar{x}_4)^2 \sum_{i=1}^n (y_{4i} - \bar{y}_4)^2}}$$

Here:

$$\bar{x}_4 = \frac{\sum_{i=1}^{11} x_{4i}}{11} = \frac{1583}{11} \approx 143.91$$

$$\bar{y}_4 = \frac{\sum_{i=1}^{11} y_{4i}}{11} = \frac{1325}{11} \approx 120.45$$

Substituting the values:

$$\sum (x_{4i} - \bar{x}_4)(y_{4i} - \bar{y}_4) \approx 10294.55$$

$$\sum (x_{4i} - \bar{x}_4)^2 \approx 11149.64$$

$$\sum (y_{4i} - \bar{y}_4)^2 \approx 9728.73$$

Thus:

$$r \approx \frac{10294.55}{\sqrt{11149.64 \times 9728.73}} \approx 0.994$$

Interpretation: The correlation coefficient of approximately 0.994 indicates a very strong positive linear relationship between income and weekly consumption expenditure.

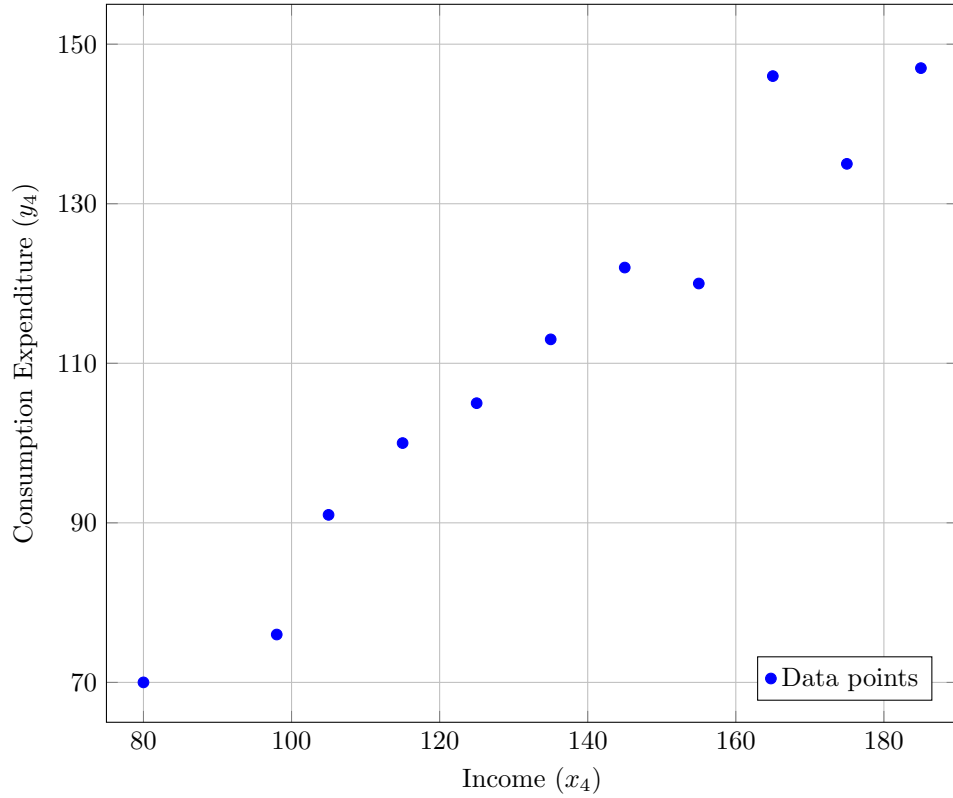


Figure 4: Scatter diagram of income vs. consumption expenditure for Dataset 4

1.5 Data Set 5

Marks (y_5) \ Attendance (x_5)	60%	70%	80%	90%	100%
60%	1	2	1	0	0
70%	1	3	3	1	0
80%	0	1	3	1	0
90%	0	0	4	4	0
100%	0	0	4	7	4

Table 5: Bivariate distribution of attendance and marks in an examination for 40 students

Each cell in the table represents the number of students.

Use the formula for the correlation coefficient as applicable.

Solution.

We assign numerical codes to the percentage values:

$$x_5 : \{60, 70, 80, 90, 100\}, \quad y_5 : \{60, 70, 80, 90, 100\}.$$

Let f_{ij} denote the frequency corresponding to (x_i, y_j) . The total frequency is:

$$N = \sum_{i=1}^5 \sum_{j=1}^5 f_{ij} = 40.$$

The marginal totals for x_5 :

$$f_{x_5} = \{4, 7, 15, 13, 1\}.$$

The marginal totals for y_5 :

$$f_{y_5} = \{4, 8, 5, 8, 15\}.$$

The means are:

$$\bar{x}_5 = \frac{\sum x_5 f_{x_5}}{N} = \frac{(60)(4) + (70)(7) + (80)(15) + (90)(13) + (100)(1)}{40} = \frac{3240}{40} = 81.$$

$$\bar{y}_5 = \frac{\sum y_5 f_{y_5}}{N} = \frac{(60)(4) + (70)(8) + (80)(5) + (90)(8) + (100)(15)}{40} = \frac{3400}{40} = 85.$$

We use:

$$r = \frac{\sum_{i=1}^5 \sum_{j=1}^5 f_{ij} (x_i - \bar{x}_5)(y_j - \bar{y}_5)}{\sqrt{\sum_{i=1}^5 f_{x_5} (x_i - \bar{x}_5)^2 \cdot \sum_{j=1}^5 f_{y_5} (y_j - \bar{y}_5)^2}}.$$

First, compute $\sum_{i=1}^5 f_{x_5} (x_i - \bar{x}_5)^2$:

$$\begin{aligned} &= (4)(60 - 81)^2 + (7)(70 - 81)^2 + (15)(80 - 81)^2 + (13)(90 - 81)^2 + (1)(100 - 81)^2 \\ &= (4)(441) + (7)(121) + (15)(1) + (13)(81) + (1)(361) \\ &= 1764 + 847 + 15 + 1053 + 361 = 4040. \end{aligned}$$

Similarly, compute $\sum_{j=1}^5 f_{y_5}(y_j - \bar{y}_5)^2$:

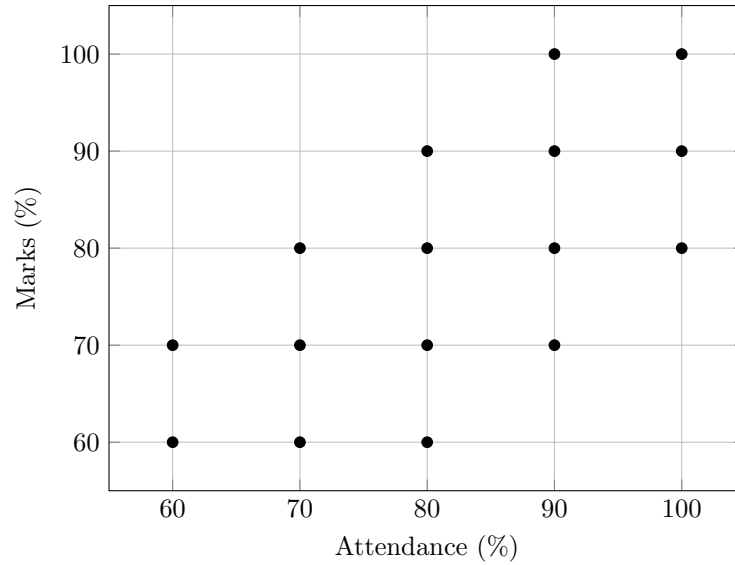
$$\begin{aligned} &= (4)(60 - 85)^2 + (8)(70 - 85)^2 + (5)(80 - 85)^2 + (8)(90 - 85)^2 + (15)(100 - 85)^2 \\ &= (4)(625) + (8)(225) + (5)(25) + (8)(25) + (15)(225) \\ &= 2500 + 1800 + 125 + 200 + 3375 = 8000. \end{aligned}$$

Next, compute $\sum_{i=1}^5 \sum_{j=1}^5 f_{ij}(x_i - \bar{x}_5)(y_j - \bar{y}_5)$:

$$\begin{aligned} &(1)(-21)(-25) + (2)(-11)(-25) + (1)(-1)(-25) + (0)(9)(-25) + (0)(19)(-25) \\ &+ (1)(-21)(-15) + (3)(-11)(-15) + (3)(-1)(-15) + (1)(9)(-15) + (0)(19)(-15) \\ &+ (0)(-21)(-5) + (1)(-11)(-5) + (3)(-1)(-5) + (1)(9)(-5) + (0)(19)(-5) \\ &+ (0)(-21)(5) + (0)(-11)(5) + (4)(-1)(5) + (4)(9)(5) + (0)(19)(5) \\ &+ (0)(-21)(15) + (0)(-11)(15) + (4)(-1)(15) + (7)(9)(15) + (4)(19)(15) \\ &= 525 + 550 + 25 + 0 + 0 + 315 + 495 + 45 - 135 + 0 + 0 + 55 + 15 - 45 + 0 + 0 + 0 - 20 + 180 + 0 + 0 + 0 - 60 + 945 + 1140 \\ &= 4260. \end{aligned}$$

Thus:

$$r = \frac{4260}{\sqrt{(4040)(8000)}} \approx \frac{4260}{5682.57} \approx 0.75.$$



Comment: The correlation coefficient $r \approx 0.75$ indicates a strong positive linear relationship between attendance and marks. Higher attendance generally corresponds to higher marks in the examination.